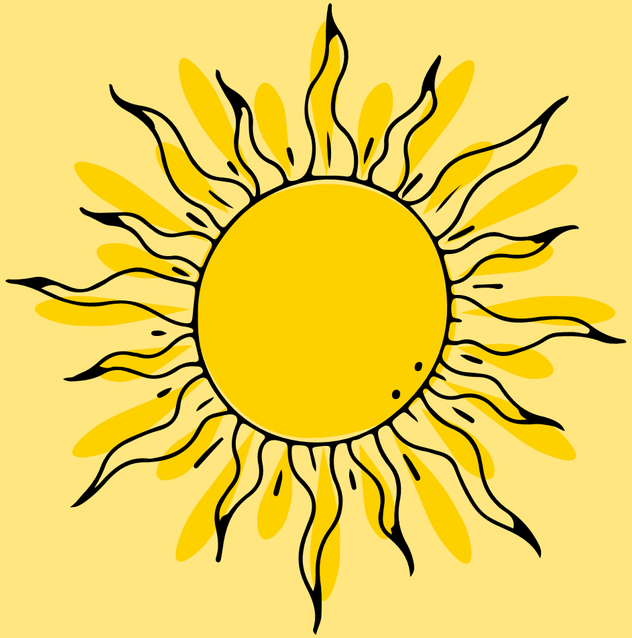
SYDNEY
REIS

APRIL
2026

ForecastFits



ALICIA
LAVENDER

MILIJANA
STOJANOVIC

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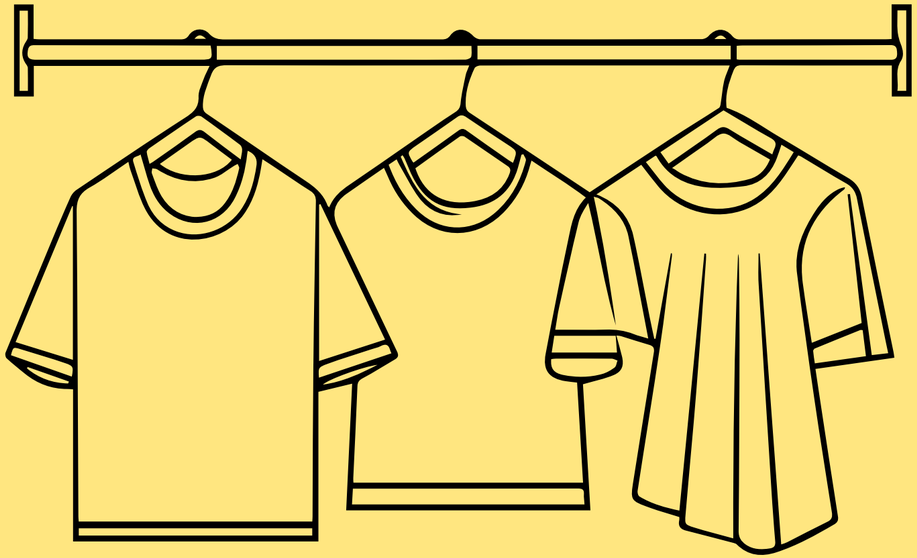

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Picking An Outfit

- What do I want to wear?
- What goes with what?
- Is this weather appropriate?
- Is it appropriate for the occasion?

Ideation Process

STYLEBOOK®

- Photograph and store items in closet
- Manually create outfits
- No automated outfit generation based on weather or occasion.

ForecastFits

Weather-based outfit recommendations

Occasion-based outfit recommendations

Fully automated outfit generation

Completely free to use

Cladwell

- Offers AI outfit suggestions
- No outift generated based on weather
- Requires a subscription for full features

UML Overview

UserProfile

- Stores user information (username, password, zip code)
- Owns the user's closet
- Handles login/logout and file saving/loading

AuthSystem

- Handles user authentication
- Interacts with UserProfile for verification
- Creates and reads user profile .txt files

Clothing

- Abstract class representing all clothing items
- Stores shared attributes (color, warmth, dressiness)
- Parent class for all clothing types

Closet

- Stores clothing items and saved outfits
- Handles adding, removing, and displaying items
- Generates outfits based on weather and dressiness

Key Relationships

Inheritance (Is-a)

- All clothing types inherit from Clothing
- e.g. Top ——▷ Clothing

Aggregation (Has-a)

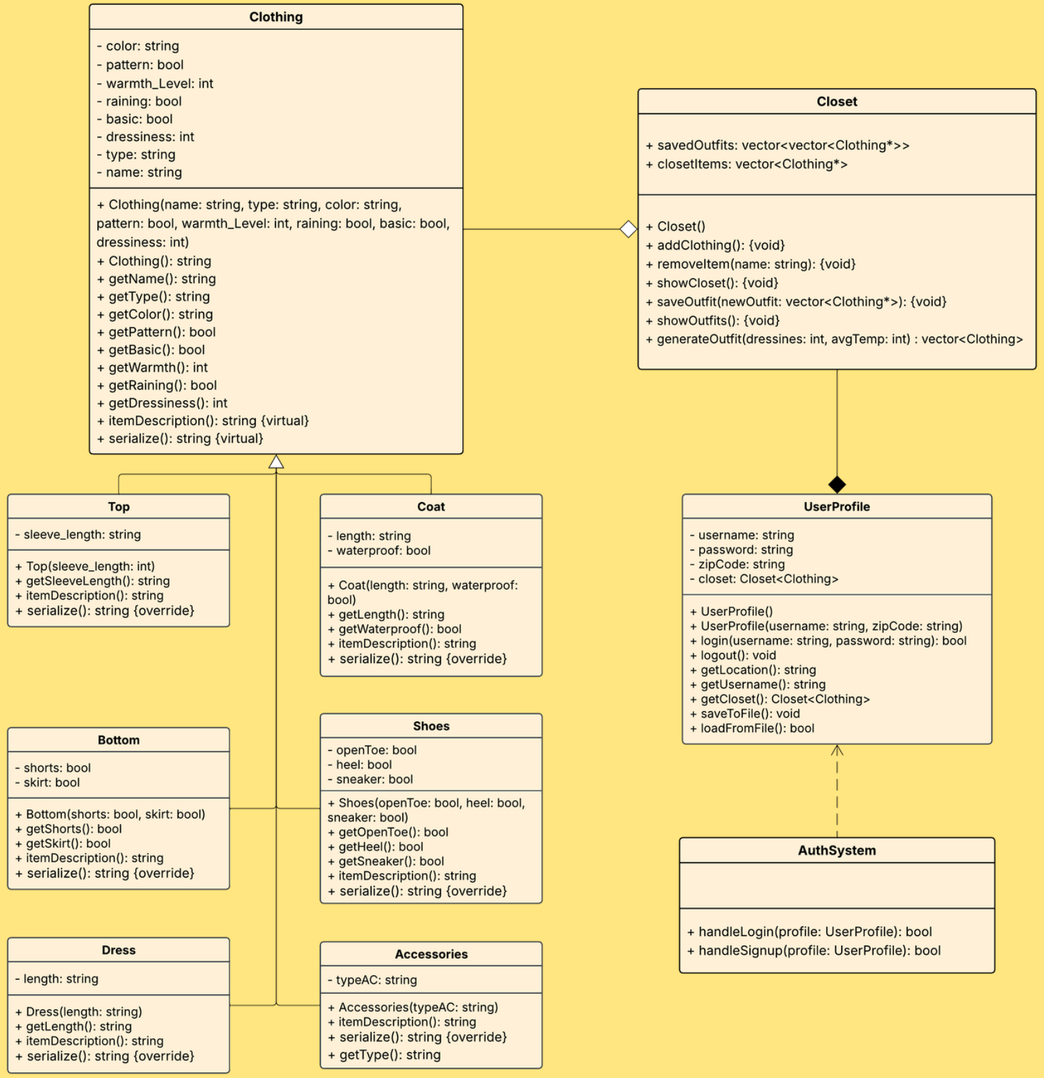
- Closet ◇———— Clothing

Composition (Has-a)

- UserProfile ◆———— Closet

Dependency (Uses)

- AuthSystem - - - - > UserProfile



System Architecture

The system follows a modular object-oriented architecture, separating **authentication**, **closet management**, and **clothing data**.

Why?

This design enables:

- **Encapsulation**
- **Inheritance**
- **Polymorphism**
- **Scalability**

Key Code Structure

Clothing.cpp

- **Each clothing item has the following shared characteristics:**
  - **name**
  - **type**
  - **color**
  - **warmth**
  - **is it patterned?**
  - **is it a basic?**
  - **dressiness**
  - **would you wear it in rain?**

Each individual type of clothing has additional attributes

ex: bottoms have:

- **is it shorts?**
- **is it a skirt?**

- **These additional attributes are used in the program for more accurate item descriptions (virtual method), and also by the generate outfit function**

Logic

Closet.h Methods

- When clothing is added it is added to a vector containing all the items a user has in there "closet"
- When removing an item, it is simply popped from the vector by name
- Viewing closet simply outputs all of the outfits stored in the vector
- If a user chooses to save a generated outfit that outfit is added to a new vector of outfits
- Main loop that users can navigate to add clothing, remove, clothing etc

Generate Outfit

1. First a user's closet is sorted into new vectors (ex: all tops put in top vector)
2. Items that are not the user's desired level of dressy are popped from the vector
3. Items that are not the user's desired level of warmth are popped from the vector
4. From the remaining items in the vectors (if there are sufficient items) an outfit is randomly generated
5. Under certain conditions the outfit is tweaked (ex: if it is <50 degrees a coat is added)

DEMO

HOW EASY WAS IT TO _____ ON A SCALE OF 1 (EXTREMELY DIFFICULT) TO 5 (EXTREMELY INTUITIVE)?

4.8

	INTERNAL	EXTERNAL	ATHLETE	CS MAJOR	OVERALL
CREATE A PROFILE	5	5	5	5	5
VIEW CLOSET	5	5	5	5	5
ADD/REMOVE CLOTHING	5	5	4.5	5	4.88
GENERATE AN OUTFIT	5	5	4	5	4.75
SAVE OUTFITS	5	5	5	3	4.5
USE	5	4	5	5	4.75

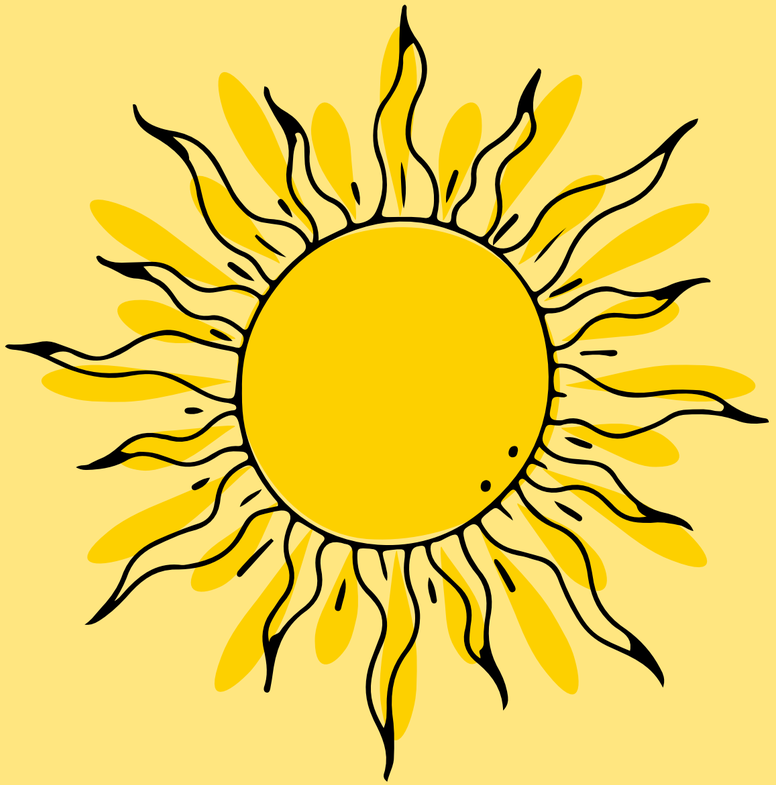
Final Reflections and Future Steps

Lessons Learned:

- Complexity of designing style-based algorithms
- Converting from text-based to WASM

Potential Improvements:

- Generate outfit algorithm
- Implementing images in the design



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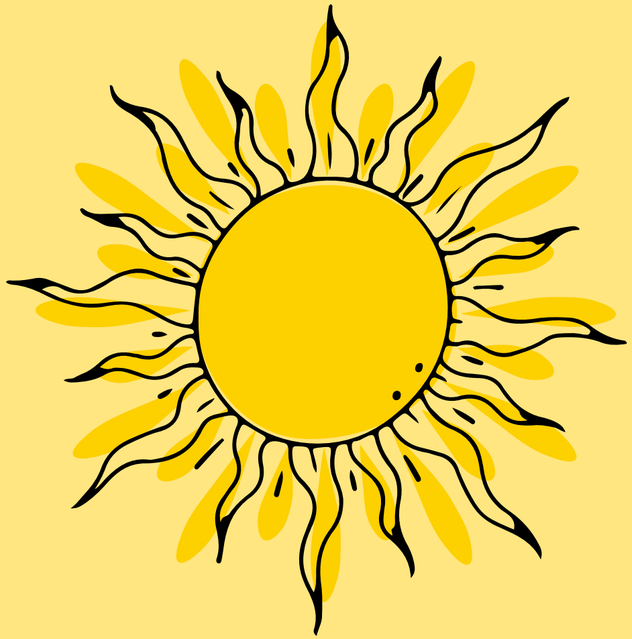
THANKS



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